

Virtualization: Containers (Part-2)

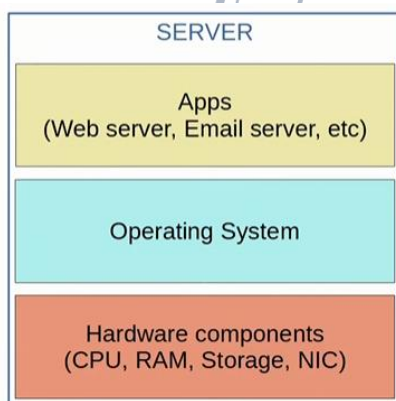
- Topics covered:
 - Review of Virtual Machines
 - Containers (ie. Docker)
 - Virtual Machine Vs Containers

Review of Virtual Machines

Applications on a Server Without Virtualization

A traditional server runs a single operating system, and all applications run on top of that OS.

Architecture



Key Points

- A single operating system is installed, such as Windows Server or Red Hat Linux.
- All applications run on top of that operating system.
- Applications are not isolated from each other.
- A problem in one application can affect all other applications.
- One solution is to use a separate physical server for each application, but this is extremely cost-inefficient.

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Virtual Machines (VMs)

Virtual Machines solve the isolation problem by allowing multiple operating systems to run on a single physical server.

Key Points

- Multiple operating systems run on one physical server.
- Applications are installed on those operating systems.
- Hardware resources are managed by a hypervisor.
- Since all VMs share the same physical hardware, the hypervisor allocates resources appropriately.

Hypervisors

A hypervisor manages and allocates hardware resources to virtual machines.

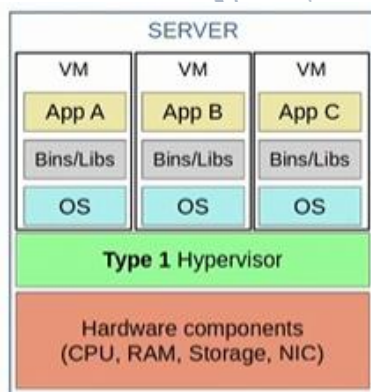
There are two main types:

1. Type 1 Hypervisor (Native / Bare-Metal)
2. Type 2 Hypervisor (Hosted)

Type 1 Hypervisor (Native / Bare-Metal)

Type 1 hypervisors run directly on hardware.

Architecture



Type 1 (Native / Bare-Metal)

Usage

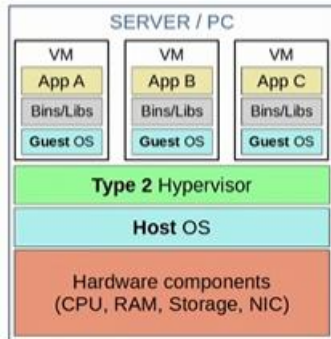
1. Widely used in data center environments.
2. Enterprise servers running VMs typically use Type 1 hypervisors.

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Type 2 Hypervisor (Hosted)

Type 2 hypervisors run on top of a Host OS.

Architecture



Type 2 (Hosted)

Usage

- Commonly used on personal devices.
- Example:
 - Windows installed on PC hardware.
 - VMware Workstation runs on Windows.
 - Cisco Modeling Labs (CML) runs as a VM inside VMware Workstation.

VM Characteristics

Operating Systems

Each VM can run the same or a different operating system.

Examples:

- One VM running Windows.
- One VM running Linux.
- One VM running macOS.

Binaries and Libraries

The gray blocks labeled **Bins/Libs** represent:

- Software libraries
- Services required by the applications inside the VM

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Benefits of VMs

1. Applications run in isolated environments.
2. Problems in one VM do not affect other VMs.
3. Problems in one Guest OS do not affect other Guest OSs.
4. VMs are easy to: Create, Delete, Move
5. A VM can be saved and moved between physical servers.

Summary: Virtualization with VMs enables much more efficient use of server hardware and is widely used today.

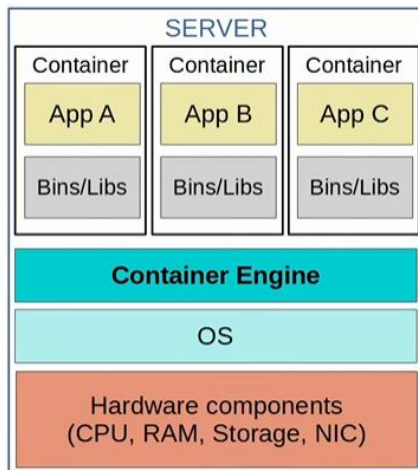


Containers

What are Containers?

Containers are a newer virtualization technology that is increasingly used.

Container Architecture



What is a Container?

Containers are software packages that contain:

- An application
- All dependencies required by that application
- Binaries
- Libraries

Although multiple applications can run in one container, containers are generally used with:

One Container = One App

Container Engine

Containers run on a Container Engine.

Example: Docker Engine

Docker Engine is the most popular container engine.

The container engine runs on a host operating system, usually Linux, which runs on the hardware.

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Major Difference Between VMs and Containers

Virtual Machines

VM

└─ App

└─ Binaries/Libraries

└─ OS

Containers

Container

└─ App

└─ Binaries/Libraries

Key Difference

1. Every VM runs its own operating system.
2. Containers do not run an operating system inside each container.
3. All containers share the host operating system.

This architectural difference creates all of the major costs and benefits when comparing VMs and containers.

Container Orchestrators

A container orchestrator is a software platform that automates:

- Deployment
- Management
- Scaling
- Other container operations

Examples: Kubernetes, Docker Swarm

Why Orchestration Is Needed

Manual management is possible for small numbers of containers.

However, large-scale systems may require thousands of containers.

Managing that many containers manually is not realistic.

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Microservice Architecture

Microservice Architecture divides a large solution into smaller parts called microservices.

Instead of: One Monolithic Application

There can be: Many Microservices

These microservices run in containers and can be orchestrated using platforms such as Kubernetes.



Virtual Machines VS Containers

Comparing VMs and Containers

The differences come from one architectural fact: VMs run their own OS.

Containers share a common OS.

Boot Time

Virtual Machines: Can take minutes to boot.

Containers: Can boot in milliseconds.

Result: Containers are much more agile.

If a container crashes, another can start very quickly and take over.

Disk Space

Virtual Machines: Often require tens of gigabytes.

Containers: Usually require tens of megabytes.

Reason:

1. VMs include an operating system.
2. Containers contain only the application and its dependencies.

CPU and RAM Usage

Virtual Machines

- Use more CPU resources.
- Use more RAM resources.

Containers

- Use fewer CPU resources.
- Use fewer RAM resources.

Reason: Again, each VM includes its own operating system while containers do not.

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Portability

Virtual Machines:

Can move between physical systems running the same hypervisor.

Containers:

1. Even more portable.
2. Smaller in size.
3. Faster to start.
4. Docker containers can run on nearly any container service.

Isolation

Virtual Machines:

- More isolated.
- Each VM has its own operating system.
- Problems in one OS do not affect other VMs.
- Isolation provides security benefits.

Containers:

- Share the same operating system.
- If the shared OS crashes, all containers are affected.

Summary

1. VMs allow multiple operating systems to run on a single physical server.
2. Hypervisors manage hardware resources for VMs.
3. Type 1 hypervisors run directly on hardware.
4. Type 2 hypervisors run on a Host OS.
5. Containers package applications and dependencies without including a separate OS.
6. Containers run on a container engine such as Docker Engine.
7. Kubernetes and Docker Swarm are examples of container orchestrators.
8. Containers boot faster, use fewer resources, and require less storage than VMs.
9. VMs provide stronger isolation because each VM runs its own operating system.
10. Containers are becoming increasingly popular with microservices, automation, and DevOps.
11. VMs remain widely used and continue to be an important virtualization technology.